

Visualizing Multilevel Models

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1 Introduction

An evolving set of notes on visualizing results from multilevel models.

The examples below use the `simulated_multilevel_data.dta` file from my text book on *Multilevel Thinking*. Here is a [link to download the data](#).

This document relies on the extraordinary `Statamarkdown` library (Hemken 2023).

2 Organizing Questions

Try to think about some of the advantages and disadvantages of different approaches to visualizing multilevel models. In multilevel models, we don't want to just *control for* variation, but to start to *explore* the variation. Put concretely:

- Some approaches use *dots*. Some approaches use *lines*. Some approaches use *dots and lines*.
- Some approaches use the *raw unadjusted* data. Other approaches use *adjusted or model predicted* data.
- Some approaches attempt to show the *Level 2 specific regression lines*; some approaches only show an *average regression line*.
- What approaches might work well with *large numbers* of Level 2 units? What approaches might work well with *smaller numbers* of Level 2 units?

What approach(es) do you prefer?

3 Setup

I am not terrifically fond of the default `s2color` graph scheme in earlier versions of Stata. Here I make use of the `michigan` graph scheme available at: <https://agrogan1.github.io/Stata/michigan-graph-scheme/>.

```
set scheme michigan
```

Stata's `stcolor` scheme—available in newer versions of Stata—would also be an option as would be Asjad Naqvi's incredible `schemepack`: <https://github.com/asjadnaqvi/stata-schemepack>.

Throughout the tutorial, I make frequent use of the `mcolor(%30)` option to add some visual interest to scatterplots by adding transparency to the markers.

4 Get Data

```
use "simulated_multilevel_data.dta", clear
```

5 Scatterplots (twoway scatter y x)

```
twoway scatter outcome warmth, mcolor(%30)  
graph export myscatter.png, width(1500) replace
```

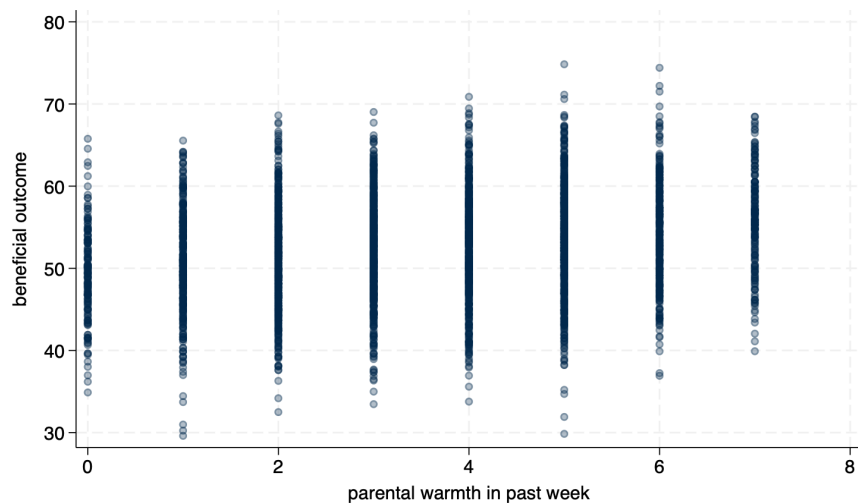


Figure 1: Scatterplot

6 Simple Linear Fit (twoway lfit y x)

```
twoway lfit outcome warmth  
  
graph export mylinear.png, width(1500) replace
```

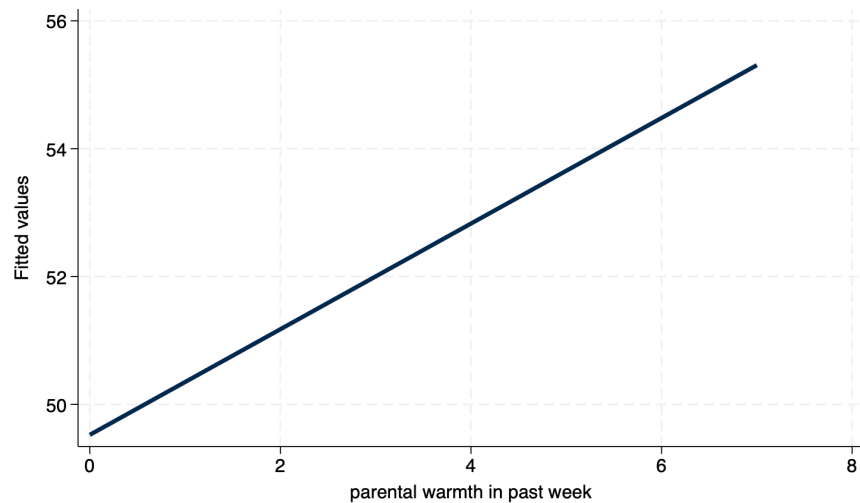


Figure 2: Linear Fit

7 Linear Fit With Confidence Interval (twoway lfitci y x)

```
twoway lfitci outcome warmth  
  
graph export mylfitci.png, width(1500) replace
```

8 Combine Scatterplot and Linear Fit (twoway (scatter y x) (lfit y x))

```
twoway (scatter outcome warmth, mcolor(%30)) (lfit outcome warmth)  
  
graph export myscatterlinear.png, width(1500) replace
```

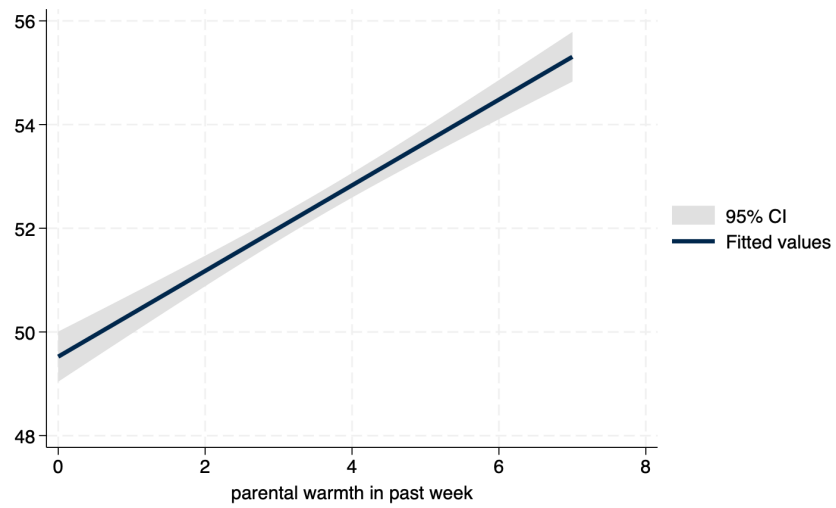


Figure 3: Linear Fit With Confidence Interval

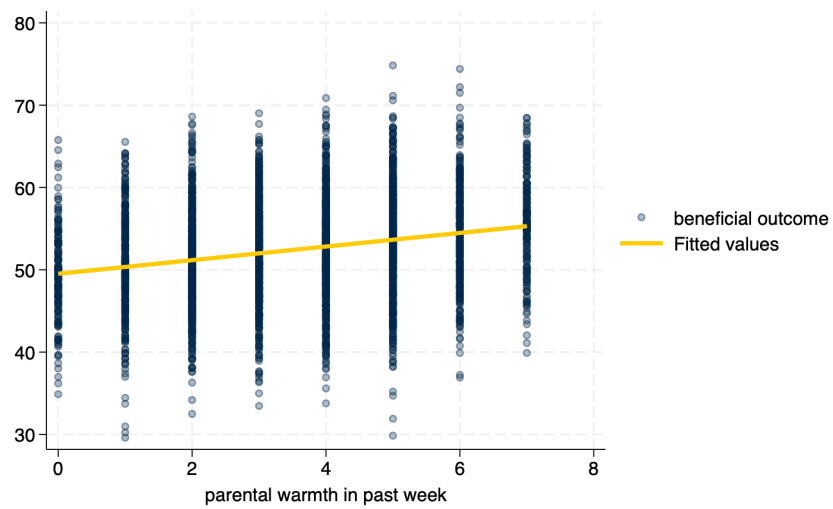


Figure 4: Scatterplot and Linear Fit

9 Spaghetti Plots (`spagplot y x, id(group)`)

```
spagplot outcome warmth, id(country)
graph export myspaghetti.png, width(1500) replace
```

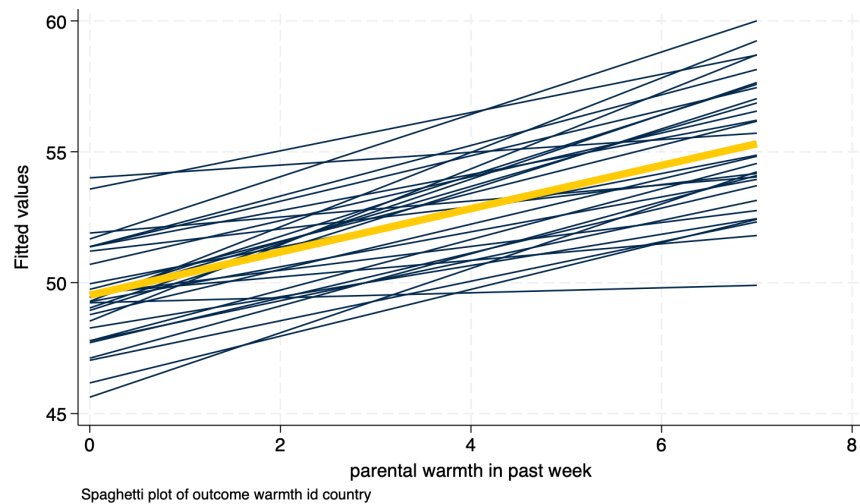


Figure 5: Spaghetti Plot

10 Small Multiples (`twoway y x, by(group)`)

Small Multiples, showing a separate graph for each group in the data, are an increasingly popular data visualization technique. Below, I build a small multiples graph using the `by` option in Stata. I use the `aspect` option to adjust the *aspect ratio* of the graph for better visual presentation.

```
twoway (scatter outcome warmth, mcolor(%30)) ///
(lfit outcome warmth), ///
by(country) aspect(1)

graph export mysmallmultiples.png, width(1500) replace
```

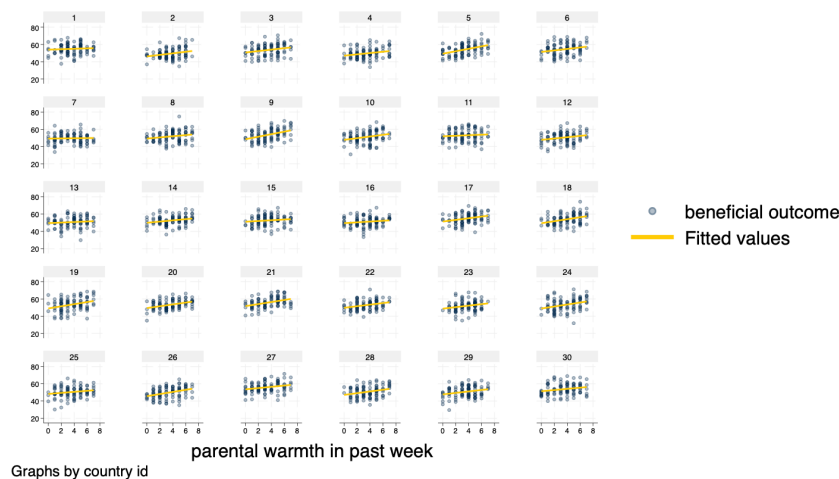


Figure 6: Small Multiples

11 Small Multiples With A Random Sample

At times, we may have *too many* Level 2 units to effectively display them on a *spaghetti plot*, or using *small multiples*. If this is the case, we may need to *randomly sample* **Level 2** units. This can be difficult to accomplish as our standard `sample` command operates on each row, or on Level 1 units.

We can accomplish random sampling at Level 2, with a little bit of code.

```
set seed 3846 // random seed for reproducibility

gen randomid = runiform() // generate a random id variable

* by country (i.e. by Level 2 unit) replace the randomid
* with the first randomid for that country (Level 2 unit)
* so that every person in that country has the same random id

bysort country: replace randomid = randomid[1]

summarize randomid // descriptive statistics for random id

twoway (scatter outcome warmth, mcolor(%30)) /// scatterplot
(lfit outcome warmth) /// linear fit
if randomid < .5, /// only use a subset of randomids
```

```
by(country) aspect(1) // by country
quietly: graph export mysmallmultiples2.png, width(1500) replace
```

(2970 real changes made)

Variable	Obs	Mean	Std. dev.	Min	Max
-----+-----					
randomid	3,000	.6174022	.2374704	.0733026	.9657055

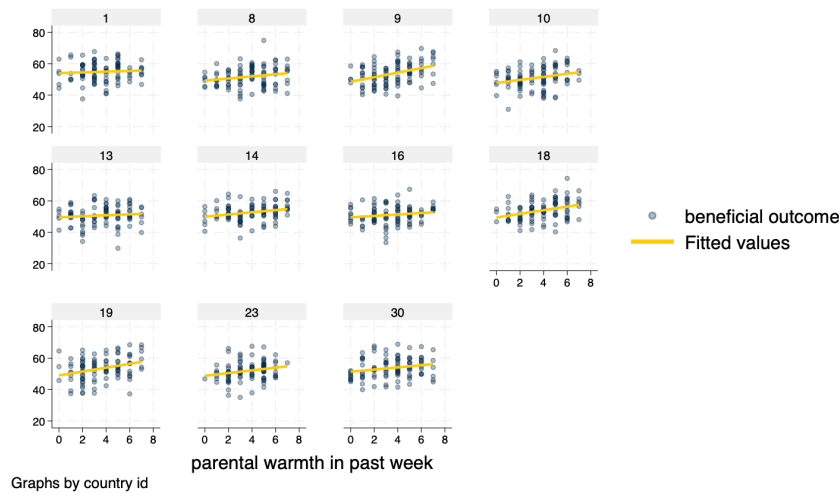


Figure 7: Small Multiples With A Random Sample Of Countries

12 Multivariate (Predicted) Relationships

💡 Multivariate vs. Bivariate Graphs

A sometimes unacknowledged point is that graphs—unless we take steps to correct this—reflect *unadjusted*, or *bivariate* associations. We may sometimes wish to develop a graphs that reflect the *adjusted* or *predicted* estimates from our models.

12.1 Using Predicted Values (predict)

`predict` generates a predicted value for *every observation in the data*.

Prediction Requires Careful Thinking

In multilevel models, *prediction* is a complex question. Prediction may—or may not—incorporate the information from the random effects. The procedures below outline graphs that incorporate predictions using the random effects, by using the `predict ...`, `fitted` syntax.

12.1.1 Estimate The Model

```
mixed outcome warmth physical_punishment i.intervention || country: // estimate MLM
```

Performing EM optimization ...

Performing gradient-based optimization:

Iteration 0: Log likelihood = -9628.1621

Iteration 1: Log likelihood = -9628.1621

Computing standard errors ...

Mixed-effects ML regression

Group variable: country

Number of obs = 3,000

Number of groups = 30

Obs per group:

min = 100

avg = 100.0

max = 100

Wald chi2(3) = 370.90

Prob > chi2 = 0.0000

Log likelihood = -9628.1621

outcome	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
warmth	.8330937	.0574809	14.49	0.000	.7204332	.9457543
physical_punis~t	-.9937819	.0798493	-12.45	0.000	-1.150284	-
.8372801						
1.intervention	.6406043	.2175496	2.94	0.003	.214215	1.066994
_cons	51.65238	.4664841	110.73	0.000	50.73809	52.56668

Random-effects parameters		Estimate	Std. err.	[95% conf. interval]	
country: Identity					
	var(_cons)	3.371762	.9613269	1.928279	5.895816
	var(Residual)	35.0675	.910002	33.32853	36.89721
LR test vs. linear model: chibar2(01) = 204.14 Prob >= chibar2 = 0.0000					

12.1.2 Generate Predicted Values

```
predict outcome_hat, fitted // predict yhat (`fitted` uses fixed AND random effects)
```

12.1.3 Graph With twoway Syntax

```
twoway (scatter outcome_hat warmth, mcolor(%30)) (lfit outcome_hat warmth)
graph export mypredictedvalues.png, width(1500) replace

twoway (lfit outcome_hat warmth)
graph export mypredictedvalues2.png, width(1500) replace
```

12.1.4 Spaghetti Plot With Predicted Values

```
spagplot outcome_hat warmth, id(country)
graph export myspaghetti2.png, width(1500) replace
```

12.2 margins and marginsplot

In contrast to `predict`, which generates a predicted value for *every observation in the data*, `margins`¹ generates predicted values at *specific values of certain variables*.

¹While it is some ways sensible, the term `margins` has always seemed unintuitive to me, so I prefer to think of `margins` as a synonym for *predicted values*.

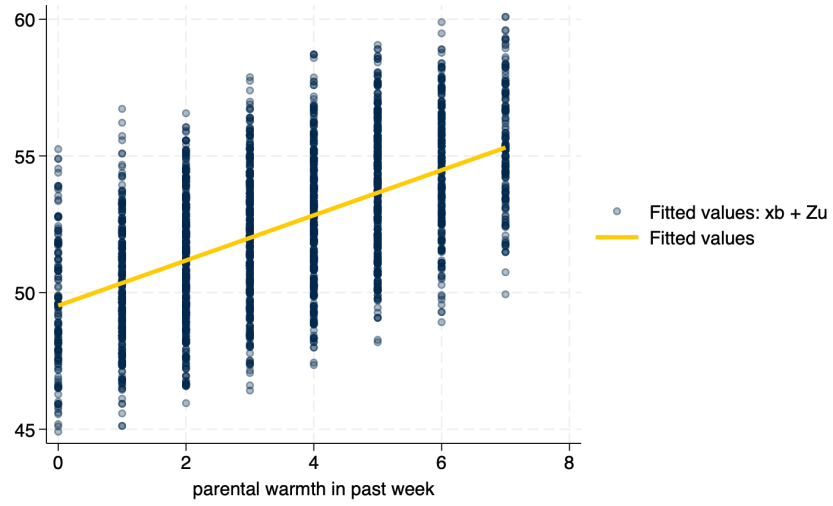


Figure 8: Predicted Values From `predict`

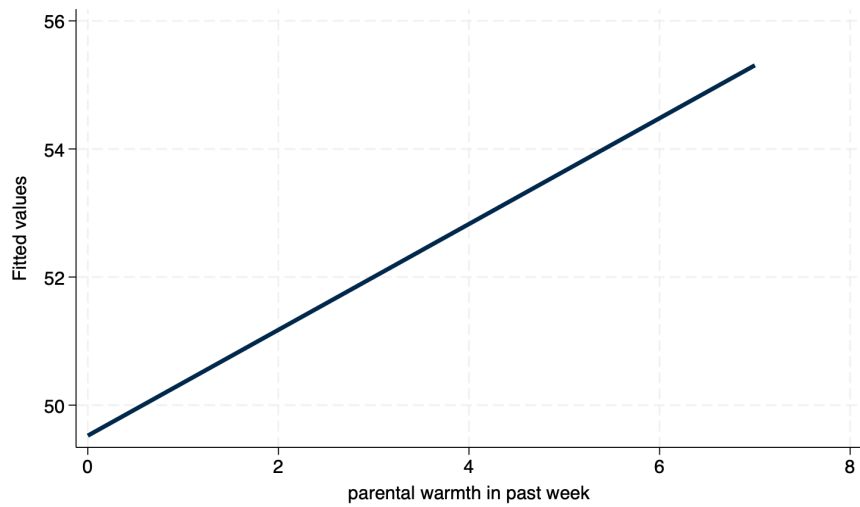


Figure 9: Predicted Values From `predict` With Only Linear Fit

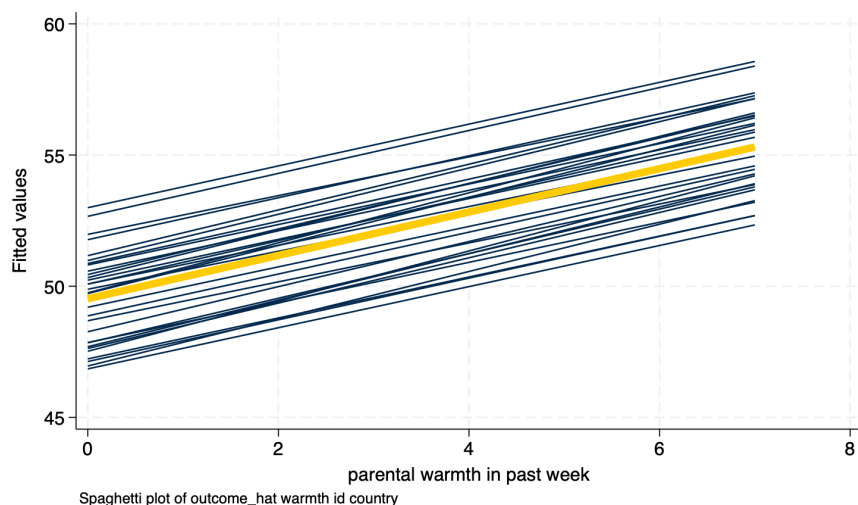


Figure 10: Spaghetti Plot With Predicted Values

12.2.1 Estimate The Model

```
mixed outcome warmth physical_punishment i.intervention || country: // estimate MLM
```

Performing EM optimization ...

Performing gradient-based optimization:

Iteration 0: Log likelihood = -9628.1621

Iteration 1: Log likelihood = -9628.1621

Computing standard errors ...

Mixed-effects ML regression

Group variable: country

Number of obs = 3,000

Number of groups = 30

Obs per group:

min = 100

avg = 100.0

max = 100

Wald chi2(3) = 370.90

Prob > chi2 = 0.0000

Log likelihood = -9628.1621

outcome	Coefficient	Std. err.	z	P> z	[95% conf. interval]
---------	-------------	-----------	---	------	----------------------

	warmth		.8330937	.0574809	14.49	0.000	.7204332	.9457543
physical_punis~t		-.9937819	.0798493	-12.45	0.000	-1.150284	-	
.8372801								
1.intervention		.6406043	.2175496	2.94	0.003	.214215	1.066994	
_cons		51.65238	.4664841	110.73	0.000	50.73809	52.56668	

Random-effects parameters		Estimate	Std. err.	[95% conf. interval]	
country: Identity					
var(_cons)		3.371762	.9613269	1.928279	5.895816
var(Residual)		35.0675	.910002	33.32853	36.89721

LR test vs. linear model: chibar2(01) = 204.14 Prob >= chibar2 = 0.0000

12.2.2 Generate Predicted Values *At Specified Values* With margins

```
* predicted values ACROSS values of `intervention`
* AT specific values of `warmth`

margins intervention, at(warmth = (1 2 3 4 5 6 7)) // predictive margins
```

Predictive margins

Number of obs = 3,000

Expression: Linear prediction, fixed portion, predict()

```
1._at: warmth = 1
2._at: warmth = 2
3._at: warmth = 3
4._at: warmth = 4
5._at: warmth = 5
6._at: warmth = 6
7._at: warmth = 7
```

		Delta-method				
		Margin	std. err.	z	P> z	[95% conf. interval]

_at#intervention							
1 0		50.02222	.3966755	126.10	0.000	49.24475	50.79969
1 1		50.66283	.3955286	128.09	0.000	49.88761	51.43805
2 0		50.85532	.3788571	134.23	0.000	50.11277	51.59786
2 1		51.49592	.3789096	135.91	0.000	50.75327	52.23857
3 0		51.68841	.3692182	139.99	0.000	50.96476	52.41207
3 1		52.32902	.370554	141.22	0.000	51.60274	53.05529
4 0		52.52151	.3684014	142.57	0.000	51.79945	53.24356
4 1		53.16211	.3710204	143.29	0.000	52.43492	53.8893
5 0		53.3546	.376464	141.73	0.000	52.61674	54.09246
5 1		53.9952	.3802764	141.99	0.000	53.24988	54.74053
6 0		54.18769	.3928599	137.93	0.000	53.4177	54.95768
6 1		54.8283	.3977088	137.86	0.000	54.0488	55.60779
7 0		55.02079	.4166062	132.07	0.000	54.20425	55.83732
7 1		55.66139	.4223062	131.80	0.000	54.83369	56.4891

12.2.3 Graph With marginsplot

```
marginsplot // plot of predicted values
graph export mymarginsplot.png, width(1500) replace
```

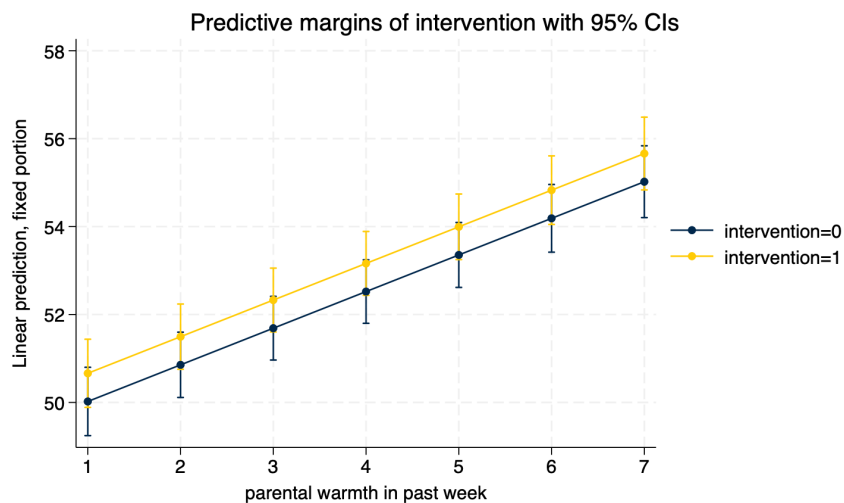


Figure 11: Predicted Values From margins and marginsplot

13 Scatterplot With Linear Fit and Marginal Density Plots (twoway ...)

As another possibility, we may wish to show more of the variation, by showing the variation in the *independent* variable and the *dependent* variable along with a *scatterplot* and *linear fit*. This is a complex graph and requires a little bit of manual programming in Stata.

i binscatterhist

You could also investigate the user written program `binscatterhist` (`ssc install binscatterhist`) which produces a similar looking graph, and automates much of this work.

13.1 Manually Generate The Densities To Plot Them Below (`kdensity ...`)

i Note

We generate the density for *warmth* at only a few points (`n(8)`) since this variable has relatively few categories.

```
kdensity warmth, generate(warmth_x warmth_d) n(8) // manually generate outcome densities
kdensity outcome, generate(outcome_y outcome_d) // manually generate outcome densities
```

13.2 Rescale The Densities So They Plot Well

i Note

You may have to experiment with the scaling and moving factors.

```
replace warmth_d = 100 * warmth_d // rescale the density so it plots well
replace outcome_d = 5 * outcome_d - .5 // rescale AND MOVE the density so it plots well
label variable outcome_y "density: beneficial outcome" // relabel y variable
```

(8 real changes made)

(50 real changes made)

13.3 Make The Graph (twoway ...)

i Note

You may have to experiment with whether scatterplots or line plots work best for displaying the x and y densities.

```
twoway (scatter outcome warmth, mcolor(%10)) /// scatterplot w some transparency
(lfit outcome warmth) /// linear fit
(line warmth_d warmth_x) /// line plot of x density
(line outcome_y outcome_d), /// line plot of y density (note flipped order)
title("Outcome by Warmth") /// title
ytitle("beneficial outcome") /// manual ytitle
xtitle("parental warmth") /// manual xtitle
legend(position(6) rows(2) ) /// legend at bottom; 2 rows
xlabel(0 1 2 3 4 5 6 7) /// manual x labels
name(mynewscatter, replace)

graph export mynewscatter.png, width(1500) replace
```

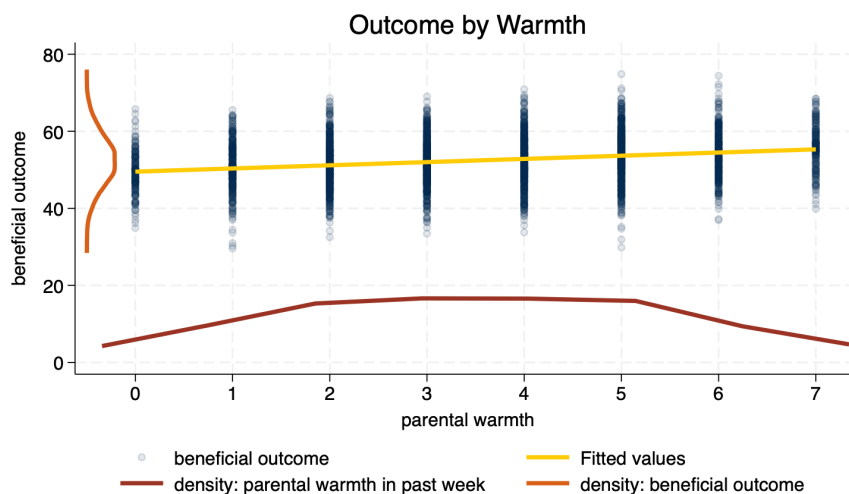


Figure 12: Scatterplot and Linear Fit With Marginal Density Plots

References

Hemken, Doug. 2023. *Statamarkdown: 'Stata' Markdown*. <https://CRAN.R-project.org/package=Statamarkdown>.